System Requirements Review

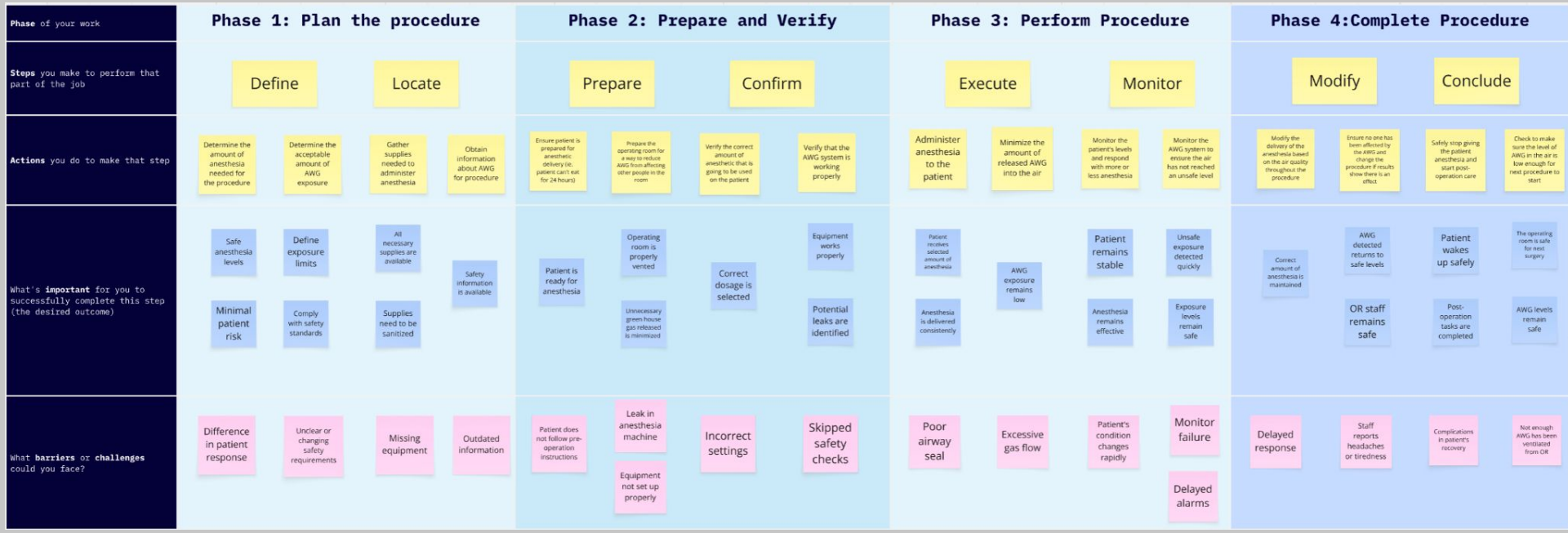
Kids OR + Public Invention: Anesthetic Waste Gas Management

Presenting: Brooke Struyk, Rachel Mensah, Kaayla Osborne, Sarah Krishnamoorthy

Problem Statement

“A way to reduce healthcare personnel exposure to WAG in low-resource operating rooms by developing a low-cost WAG management system.”

Job Map


Phase of your work
Steps you make to perform that part of the job
Actions you do to make that step
What's important for you to successfully complete this step (the desired outcome)
What barriers or challenges could you face?
Phase 1: Plan the procedure
Define
Locate
Determine the amount of anesthesia needed for the procedure
Determine the acceptable amount of AWG exposure
Gather supplies needed to administer anesthesia
Obtain information about AWG for procedure
Safe anesthesia levels
Define exposure limits
All necessary supplies are available
Safety information is available
Minimal patient risk
Comply with safety standards
Supplies need to be sanitized
Difference in patient response
Unclear or changing safety requirements
Missing equipment
Outdated information
Phase 2: Prepare and Verify
Prepare
Confirm
Ensure patient is prepared for anesthetic delivery (ie. patient can't eat for 24 hours)
Prepare the operating room for a way to reduce AWG from affecting other people in the room
Verify the correct amount of anesthetic that is going to be used on the patient
Verify that the AWG system is working properly
Patient is ready for anesthesia
Operating room is properly vented
Unnecessary green house gas released is minimized
Correct dosage is selected
Equipment works properly
Potential leaks are identified
Patient does not follow pre-operation instructions
Leak in anesthesia machine
Equipment not set up properly
Incorrect settings
Skipped safety checks
Phase 3: Perform Procedure
Execute
Monitor
Administer anesthesia to the patient
Minimize the amount of released AWG into the air
Monitor the patient's levels and respond with more or less anesthesia
Monitor the AWG system to ensure the air has not reached an unsafe level
Patient receives selected amount of anesthesia
Anesthesia is delivered consistently
AWG exposure remains low
Patient remains stable
Anesthesia remains effective
Unsafe exposure detected quickly
Exposure levels remain safe
Poor airway seal
Excessive gas flow
Patient's condition changes rapidly
Monitor failure
Delayed alarms
Phase 4:Complete Procedure
Modify
Conclude
Modify the delivery of the anesthesia based on the air quality throughout the procedure
Ensure no one has been affected by the AWG and change the procedure if results show there is an effect
Safely stop giving the patient anesthesia and start post-operation care
Check to make sure the level of AWG in the air is low enough for next procedure to start
Correct amount of anesthesia is maintained
AWG detected returns to safe levels
OR staff remains safe
Patient wakes up safely
Post-operation tasks are completed
The operating room is safe for next surgery
AWG levels remain safe
Delayed response
Staff reports headaches or tiredness
Complications in patient's recovery
Not enough AWG has been ventilated from OR

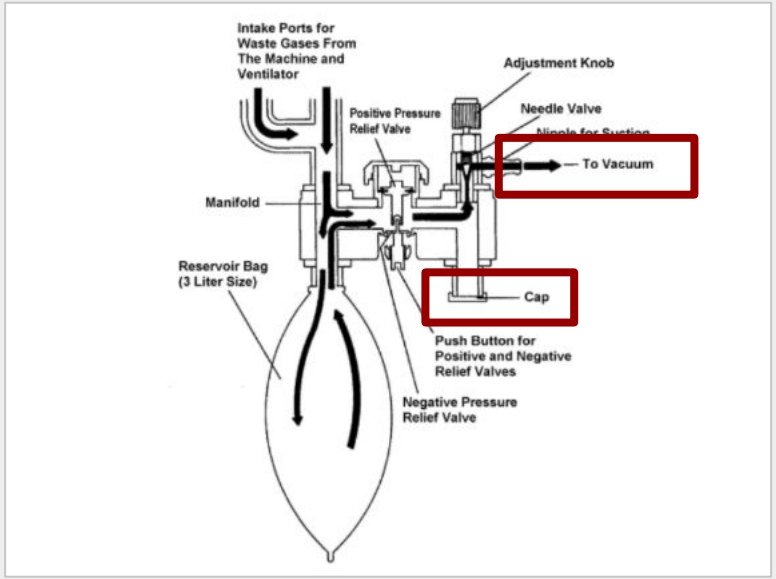
Customer Discovery Data Results

Interviewer	Interviewee	Contact	Key Insights
Brooke Struyk	Dr. Brian Struyk, MD; Pediatric Anesthesiologist and Director of Pediatric Anesthesia of the Pediatric Proton Therapy Center at the Children's Hospital of Philadelphia	struykbp@CHOP.edu	-At CHOP and globally, anesthesiologists are talking about reducing the environmental effects of AWG -Low-flow is the recommended way the anesthesiologist can reduce the about of AWG entering the air -When there is high-flow after a period of 15 minutes, the anesthesia machine will send a message on the notification bar. This bar is checked regularly throughout surgery. A picture of what this looks like is below. -There has also been a push to use sevoflurane over Nitrous oxide; however, in low resource countries nitrous oxide is the most common anesthetic because it is cheap. -Anesthesiologists typically ensure prior to the surgery that the WAGD system has been turned on and it stays on the entire day. -Anesthesiology technicians are the main maintenance team for leaks or issues. If the issue is bigger, BMET would be brought in. -The final say on a new technology goes more towards the BMET, and maintenance department is they can account for the new machine. -The picture above shows the WAGD device that is being used at CHOP. The white box is a CO2 scrubber, and the flapper valve on the right has a max and min dial to show if there is enough suction for the AWG.
Kaayla Osborne	Dr. Andy Muelenear; Pediatrician and Pulmonologist	andrem1@vt.edu	-Measure out 18 feet in our space. -Possibly running a scavenging system in a place it's not equipped to run in or not changing filters -US Military has portable ORs called Deployable Medical that are 40ft modules. -Get details of what is in the OR and models of each device
Sarah Krishnamoorthy	Mrs. Ashley Hunnicutt; Executive Director, Loudoun Free Clinic	ashleyhunnicutt@loudounfreeclinic.org	- Failure does not always mean the device doesn’t work. Failure can mean the device was too expensive, difficult to install, disruptive, or difficult to maintain. - Availability of parts for replacements and access to those who know how to service the system are vital. On-site staff could be trained or troubleshoot with a set of short videos. - Early prototyping and validation are key. She emphasized that mockups and models are crucial and can save time and money. Additionally, it’s an easy way to get feedback from end users dynamically. - The team should not just focus on initial cost, but the life cycle of the product.

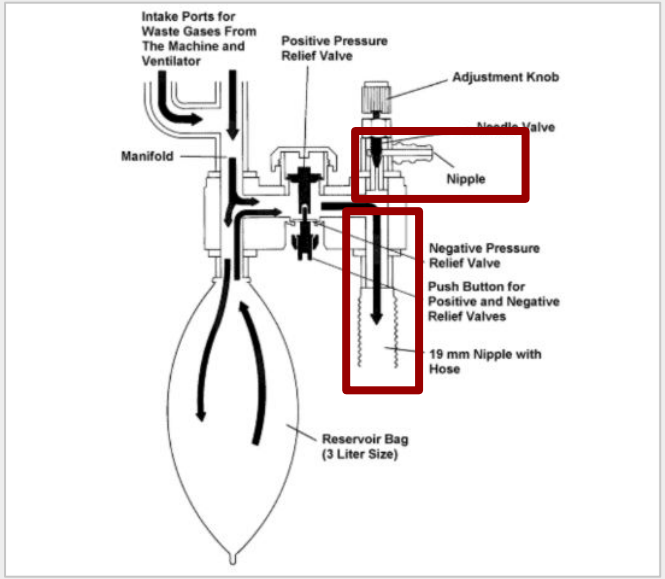
User Needs

Level 5 Needs	**Level 4 Needs**
A device that can be independently powered	A device that is modular
A device that is easy to read	A device that can be handled safely by all intended users
A system that is open- source	A device that can be cleaned regularly
A device that can meet the OR’s machine sterilization requirements	

Existing Solutions



Active



Passive

Fall Deliverables

Can we test it? Can we move it? Can we model it?

- Proof of Concept Prototype
  - Develop a proof of concept WAG disposal pathway to investigate a standalone active WAG or passive (Fan/blower for active system)
  - Gas flow rate
  - Tubing configuration
- Experimental
  - Controlled comparison of passive scavenging vs standalone active
- Computational
  - Develop a basic gas-flow model via COMSOL to compare active and passive pathways
  - Predict basic behavior, compare behavior with experimental
  - Use COMSOL model to refine prototype’s tubing and applied airflow

Team Roles

Brooke Struyk: Project Lead and Design Engineering

Kaayla Osborne: Client POC and Electronic Engineering

Rachel Mensah: Testing and Validation Engineering

Sarah Krishnamoorthy: Software and Systems Engineering

Challenges and Questions

- Rethinking project direction

- COMSOL

- Selecting computational model

Supplemental

Table 001-5 Design Control Matrix

No.	User Need	No.	Design Input	Values and Units	No.	Design Output	No.	Verification	No.	Validation	Priority	Rationale	Source
Performance and Physical Requirements													
UN-1	A device that can be independently powered.	DI-1			DO-1		VER-1		VAL-1		5	The OR itself uses a lot of power for the machines already there. Most of these countries that host the OR do not have the money or means to use more electricity.	Dr. Robert Read
		DI-2			DO-2		VER-2		VAL-2				
UN-2	A device that is modular.	DI-3			DO-3		VER-3		VAL-3		4	We want to be able to detect the gas continuously, but for the fan activation to be modular, so the fan isn't taking excessive power in large quantities so the battery can last longer.	Dr. Robert Read
		DI-4			DO-4		VER-4		VAL-4				
UN-3	A device that can be transported easily.	DI-5			DO-5		VER-5		VAL-5		3	Our client travels frequently and needs to be able to pack up the device when they change cities or countries.	Dr. Robert Read
UN-4	A device that is easy to read.	DI-6			DO-6		VER-6		VAL-6		5	Anesthesiologists must focus on ensuring the patient is safe and needs to be able to quickly view if the waste gas levels are too high in seconds.	Dr. Robert Read
Human Factors													
UN-5	A device that can be handled safely by all intended users.	DI-11									4	Any person should be able to pack up the device, set up the device, and/or read the device in case the technicians are not available.	Dr. Robert Read
		DI-12											

Supplemental

UN-6	A device that can be assembled and disassembled easily.	DI-13									2	Our client performs around 10 surgeries per day, so they need to set up equipment quickly to start on time.	Dr. Robert Read
Environmental Requirements													
UN-7	A device that doesn't push anesthetic gas into the outside air.	DI-16									3	Anesthesiologists are looking for ways to reduce the amount of WAG from entering the Earth's atmosphere.	Dr. Brian Struyk
Manufacturing Requirements													
UN-8	A system that is open-source.	DI-19									5	What is being made must be open and accessible to the public to build, add to, and comment on.	Dr. Robert Read
Sterilization, Labeling, Packaging Requirements													
UN-9	A device that can meet the OR's machine sterilization requirements	DI-22									5	OR machines need to meet the hospital's own sterilization and machine cleaning standards. This way the device can be easily implemented into the typical OR equipment and machine cleaning procedures.	Dr. Brian Struyk
UN-10	A device that can be cleaned regularly	DI-23									4	The device needs to be easy to clean to prevent risking OR sterilization because of hard to reach or incense cleaning schedules and maintenance.	Dr. Brian Struyk
UN-11	An alarm system that is clearly labeled	DI-24									2	United States regulations require clearly labeled alarm systems, and even though other countries will also be using this system, the premise of the standard should be followed as a need.	IEC 60601-1-8